

FAHIMEH ORVATI NIA

College Station, TX , fahimehorvatinia@tamu.edu , [Google Scholar](#) , [LinkedIn](#) , [GitHub](#)

EDUCATION

Texas A&M University, College Station, TX *Jan 2025, Present*
Doctor of Philosophy in Computer Engineering
Research Areas: AI, Computer Vision, Agricultural Imaging, Graph Neural Networks

University of Notre Dame, IN, USA *Aug 2023, Jan 2025*
Master of Science in Electrical Engineering

Amirkabir University of Technology, Tehran, Iran *Sept 2016, Jan 2020*
Bachelor of Science in Electrical Engineering

WORK EXPERIENCE

Research Assistant, Texas A&M AgriLife Research, College Station, TX *Jan 2025, Present*

- Working on computer vision and AI-based analysis of agricultural multispectral imagery
- Developing methods for plant growth analysis and phenotyping using deep learning
- Integrating multispectral imaging with low-field MRI for comprehensive crop analysis

Teaching Assistant, University of Notre Dame, IN, USA *Fall 2023, Spring 2024*

- Courses: Introduction to Electrical Engineering, Signals and Systems
- Developed course materials and assisted students with assignments and projects

Research Assistant, University of Tehran, Tehran, Iran *Jan 2022, June 2023*

- Worked on machine learning for classification and clustering of musical instruments

Teaching Assistant, Amirkabir University of Technology, Tehran, Iran *Sept 2020*

- Guided students in designing control systems using MATLAB Simulink

SKILLS

Languages & APIs: Python, C++, MATLAB, LaTeX
ML & CV Libraries: PyTorch, TensorFlow, Keras, OpenCV
Developer Tools: Git, Docker
Embedded Systems: STM32, Raspberry Pi, Arduino

SELECTED PROJECTS

Traffic-Aware Pedestrian Intention Prediction (Python/PyTorch)
Introduced a traffic-aware graph neural network architecture (TA-STGCN) for predicting pedestrian intentions in dynamic traffic environments. Improved performance on PIE dataset with applications in autonomous systems and smart cities.

Agricultural Imaging and Phenotyping (Python/Deep Learning)

Developing computer vision and AI-based methods for analyzing multispectral imagery of horticultural crops. Integrating multispectral imaging with low-field magnetic resonance imaging for comprehensive plant phenotyping.

GAN-Based Anomaly Detection (Python/TensorFlow)

Designed generative adversarial networks with custom loss functions for detecting anomalies in time-series data, with applications in smart meter systems and power grids.

Neighborhood Feature Pooling for Remote Sensing (Python/PyTorch)

Proposed a novel neighborhood feature pooling approach for remote sensing image classification, improving performance on agricultural and environmental monitoring tasks.

SELECTED PUBLICATIONS

Traffic-aware pedestrian intention prediction, Orvati Nia F., Lin H., 2025, American Control Conference (ACC). [arXiv:2507.12433](#)

Neighborhood Feature Pooling for Remote Sensing Image Classification, Orvati Nia F., Peebles J., et al., 2025, WACV 2026 Workshop on Earth Observation. [arXiv:2510.25077](#)

Evaluating GAN-LSTM for Smart Meter Anomaly Detection in Power Systems, Orvati Nia F., Salehi S., Peebles J., 2026, TPEC 2026. [arXiv:2601.09701](#)

Integrating Multispectral Imaging and Low-Field Magnetic Resonance Imaging for Comprehensive Phenotyping of Horticultural Crops, McFarland A., Rossi L., Nia F., Peebles J., Svyanek A., 2025, HORTSCIENCE 60 (7)

ADDITIONAL

Peer Review: LXAI-NeurIPS 2025, IEEE Robotics and Automation Letters (RA-L), ICRA 2026, American Control Conference (ACC) 2025, Conference on Control Technology and Applications (CCTA) 2025

Student Mentoring: Chloe Tiley, Sophia Martinez-Badiillo, Michael Morse, Amran Kassaye, Nazar Oladepo

Honors: National full undergraduate scholarship in Iran, Top rank in Computational Intelligence and Communication Systems courses, Top rank in Iranian National Entrance Exam for Master's programs, Travel award \$2,000 from PlantCV